Hamdard University
Department of Computing
Final Year Project

FINAL YEAR PROJECT AUTOMATION SYSTEM (FYPAS)
(FYP-028/FA-25)

FYP-1 REPORT

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Definition of Terms, Acronyms, and Abbreviations

Table 2: Definition of Terms, Acronyms, and Abbreviations

Term	Description
FYP	Final Year Project is an academic requirement completed by undergraduate students as part of a degree program.
Admin	A system user responsible for managing users, policies, scheduling, and system configuration.
Supervisor	A faculty member assigned to guide, review, and evaluate student Final Year Projects.
Committee / Jury	A group of faculty members responsible for evaluating project proposals and defenses.
RBAC	Role-Based Access Control is a security mechanism that restricts system access based on user roles.
API	Application Programming Interface is a set of protocols that allow system components to communicate.
REST	Representational State Transfer is an architectural style used for designing web services.
JSON	JavaScript Object Notation is a lightweight format for structured data exchange.
HTTPS	Hypertext Transfer Protocol Secure is a protocol used for secure data transmission.
SSL/TLS	Encryption protocols used to secure communication between client and server.
Dashboard	A user interface that displays role-specific information and system activities.
Milestone	A defined stage in the Final Year Project lifecycle used to track progress.
Proposal Defense	A formal evaluation session where students present and defend their project.

Abstract

Final Year Projects (FYPs) are an essential academic requirement for undergraduate students; however, their management in many institutions is still handled through manual and unstructured processes. These traditional approaches often result in inefficiencies, miscommunication, and lack of transparency. This project proposes the development of a Final Year Project Automation System (FYPAS) to digitize and streamline the complete FYP lifecycle. The system aims to provide role-based access for students, supervisors, coordinators, and committee members, enabling efficient project management, communication, monitoring, and evaluation. By automating key processes and centralizing data, the proposed system seeks to reduce administrative workload, improve coordination, and ensure consistent enforcement of academic policies.

Keywords: Final Year Project, Project Automation System, Academic Management System, Role-Based Access Control, Project Monitoring, University Information System

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CHAPTER 1

INTRODUCTION

1.1 Motivation

Final Year Projects (FYPs) play a crucial role in undergraduate education by enabling students to apply theoretical knowledge to real-world problems. Despite their academic importance, the management of Final Year Projects in many universities remains largely manual and fragmented. Processes such as student registration, group formation, supervisor allocation, proposal submission, evaluations, and result compilation are often handled using spreadsheets, emails, and physical documentation.

These traditional methods lead to inefficiencies, delayed communication, lack of transparency, and increased administrative workload. Students face challenges in tracking their project status and receiving timely feedback, while supervisors and coordinators struggle to manage multiple projects simultaneously. This situation highlights the need for a centralized and automated system that can streamline FYP management.

The motivation behind developing the Final Year Project Automation System (FYPAS) is to digitize and improve the overall FYP workflow by providing a structured, transparent, and efficient platform for all stakeholders involved.

1.2 Problem Statement

The current Final Year Project (FYP) process at several academic institutions is largely unorganized and relies heavily on manual procedures. This results in operational inefficiencies, frequent miscommunication among stakeholders, and missed academic deadlines. Additionally, the absence of a centralized system prevents the consistent enforcement of institutional policies, which may lead to inconsistencies, potential biases, and unfair treatment of students.

There is a clear need for an automated system that can manage the complete FYP lifecycle in a structured manner while ensuring transparency, accountability, and compliance with academic regulations.

1.3 Goals and Objectives

The primary goal of the Final Year Project Automation System (FYPAS) is to digitize and streamline the management of Final Year Projects within a university environment.

The specific objectives of the system are as follows:

- Digitize the Final Year Project Workflow:
  Transform the traditional, manual FYP process into a fully digital system that aligns with university policies and streamlines project management, communication, and documentation.

- Implement Role-Based Access Control (RBAC):
  Ensure secure, role-specific access for all stakeholders, including students, supervisors, and coordinators, based on their responsibilities within the FYP lifecycle.

- Automate Deadline Management and Notifications:
  Enforce submission deadlines through automated reminders and notifications, helping all participants remain informed and accountable throughout the FYP process.

- Enable Document Submission, Version Control, and Progress Monitoring:
  Provide structured mechanisms for document submission with version tracking and continuous monitoring of student progress to support timely feedback and accurate performance evaluation.

1.4 Project Scope

The scope of the Final Year Project Automation System includes the design and development of a web-based and Android-based application to manage Final Year Project activities efficiently. The system supports multiple user roles, including students, supervisors, coordinators, and committee or jury members.

The system covers core functionalities such as student registration, eligibility validation, group formation, supervisor selection, project proposal submission and review, real-time communication, meeting scheduling, milestone tracking, evaluation management, result publication, and secure archival of completed projects.

Features such as integration with external third-party tools, advanced analytics, and extended automation beyond the academic workflow are considered outside the scope of the current implementation and may be addressed in future enhancements.

CHAPTER 2

RELEVANT BACKGROUND & DEFINITIONS

2.1 Relevant Background

Final Year Projects (FYPs) represent a critical component of undergraduate education, enabling students to demonstrate their technical knowledge, problem-solving abilities, and research skills. Traditionally, FYP management in academic institutions involves multiple stakeholders, including students, supervisors, coordinators, and evaluation committees. These stakeholders interact across several stages such as registration, proposal submission, supervision, evaluation, and final project archival.

In many universities, these processes are managed using manual or semi-digital methods, including emails, spreadsheets, and paper-based records. Such approaches often result in fragmented data storage, lack of transparency, delayed feedback, and difficulties in monitoring student progress. Additionally, enforcing academic policies such as supervision limits, eligibility criteria, and evaluation rules becomes challenging without centralized control.

With the increasing adoption of digital solutions in academic environments, there is a growing need for automated systems that can efficiently manage complex academic workflows. Web-based and mobile-based platforms provide accessibility, real-time communication, and centralized data management, making them suitable for handling academic processes such as Final Year Projects. The Final Year Project Automation System (FYPAS) is proposed to address these challenges by offering a structured, secure, and role-based digital solution for managing the entire FYP lifecycle.

2.2 Definitions

To ensure clarity and consistency throughout this report, the following key terms and acronyms are used:

Term	Description
FYP	Final Year Project is an academic requirement completed by undergraduate students as part of a degree program.
Admin	A system user responsible for managing users, policies, scheduling, and system configuration.
Supervisor	A faculty member assigned to guide, review, and evaluate student Final Year Projects.
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Term	Description
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CHAPTER 3

LITERATURE REVIEW & RELATED WORK

Literature Review

The management of Final Year Projects (FYP) has been a subject of increasing attention in academic institutions due to its complexity and administrative demands. Traditional FYP management systems, particularly in Pakistani universities, have historically relied on manual or semi-digital processes. These systems typically support basic functionalities such as student registration, proposal submission, supervisor assignment, and final report submission. However, their reliance on limited automation and lack of real-time interaction has restricted their effectiveness.

Studies and academic projects developed over the years indicate that early FYP Management Systems (FYPMS) focused primarily on digitizing basic workflows without addressing communication efficiency, real-time progress monitoring, or user experience. While such systems reduced paperwork, they failed to provide comprehensive automation or transparency across the full FYP lifecycle.

Recent research emphasizes the importance of role-based access control, centralized data management, automated notifications, and mobile accessibility in academic systems. These features are considered essential for ensuring accountability, improving collaboration, and supporting scalable academic workflows

Related Work

Several Final Year Project Management Systems have been proposed and implemented over time, each contributing incrementally to improving FYP automation.

The FYPMS developed at Hamdard University in 2018 by Fatima Naeem introduced a structured digital approach by supporting online registration, eligibility verification, proposal submission, supervisor selection, group formation, meeting scheduling, feedback mechanisms, and administrative control. While effective in handling core FYP processes, the system lacked advanced features such as real-time messaging, automated notifications, file version control, mobile accessibility, and analytical reporting.

The FYPMS Project of Jinnah University (2021) extended earlier models by incorporating group project management, advisor idea posting, jury or committee management, and bulk student uploads. This system improved administrative efficiency and introduced better coordination among stakeholders. However, it still did not provide real-time communication, automated alerts, or mobile-friendly access, limiting its usability in modern academic environments.

The FYPMS Project of 2024 by Asim Ali Murtaza further enhanced system capabilities by introducing visual progress and evaluation charts, improved user dashboards, and secure authentication mechanisms. Despite these improvements, the system remained limited in areas such as real-time messaging, automated notifications, file version history, mobile application support, jury management, and bulk data handling.

Although each of these systems contributed valuable features to FYP automation, none provided a fully integrated, real-time, and cross-platform solution covering all stages of the FYP lifecycle.

Gap Analysis

Based on the comparative analysis of existing FYP management systems, several critical gaps have been identified. While all reviewed systems support fundamental features such as registration, proposal submission, supervisor selection, group formation, and evaluation management, they fall short in delivering advanced automation and modern collaboration capabilities.

Key gaps observed include the absence of real-time messaging, automated notifications, and file version history, which are essential for effective communication and document management. Additionally, most systems lack mobile application or responsive access, restricting accessibility for users who rely on mobile devices.

Furthermore, features such as jury and committee management, bulk student upload, and comprehensive report generation are either partially implemented or completely missing in existing systems. Limited visualization of project progress and evaluation metrics also reduces transparency and decision-making efficiency.

The proposed Final Year Project Automation System (FYPAS) addresses these gaps by offering a comprehensive, web-based and mobile-accessible solution. It integrates real-time communication, automated notifications, file version control, visual performance dashboards, group project management, jury handling, bulk data uploads, and analytical report generation. By consolidating all FYP activities into a single platform, FYPAS enhances efficiency, transparency, and collaboration across all stakeholders, effectively overcoming the limitations identified in existing systems.

Feature Comparison Table

Features	FYPMS of Hamdard University of 2018 by Fatima Naeem	FYPMS Project of Jinnah University of 2021	FYPMS Project of 2024 by Asim Ali Murtaza	Our Proposed FYPAS Project
Online Registration & Login	✓	✓	✓	✓
Eligibility & Proposal Submission	✓	✓	✓	✓
Supervisor Selection	✓	✓	✓	✓
Advisor Posting Ideas	✓	✓	✓	✓

Group Formation Request	✓	✓	✓	✓
Meeting Scheduling	✓	✓	✓	✓
Supervisor Feedback	✓	✓	✓	✓
File Upload/Sharing	✓	✓	✓	✓
Final Report Submission	✓	✓	✓	✓
Progress Tracking	✓	✓	✓	✓
Admin Panel (User/Marks Management)	✓	✓	✓	✓
Secure Login (JWT/Hashed Passwords)	✓	✓	✓	✓
Real-Time Messaging	✗	✗	✗	✓
Automated Notifications	✗	✗	✗	✓
File Version History	✗	✗	✗	✓
Visual Progress & Evaluation Charts	✗	✗	✓	✓
Group Project Management	✗	✓	✓	✓
Mobile App or Responsive Access	✗	✗	✗	✓
Jury/Committee Management	✗	✓	✗	✓
Bulk Student Upload	✗	✓	✗	✓
Report Generation (PDF, Analytics)	✗	✗	✗	✓